INCH-POUND
A-A-50612A
December 12, 2025
SUPERSEDING
A-A-50612
December 3, 1993

COMMERCIAL ITEM DESCRIPTION

BATTERY, STORAGE (VALVE-REGULATED LEAD-ACID, ABSORBENT GLASS MAT), 12-VOLT, 50-AMPERE-HOUR

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers a 12-volt, 50-ampere-hour (Ah), valve-regulated lead-acid (VRLA) absorbent glass mat, storage battery with a low rate and deep discharge (see 6.5.4) duty cycle for use on naval surface ships.

2. SALIENT CHARACTERISTICS. The product shall be a 12-volt, 50-Ah, VRLA storage battery meeting the requirements specified below.

2.1 Dimensions. The maximum dimensions for the battery shall be:

a. Length – 10.0 inches (254 millimeters [mm])

b. Width – 7.1 inches (180 mm)

c. Height – 9.5 inches (241 mm)

2.2 Weight. The battery weight shall be 40 pounds (18 kilograms) maximum when filled.

2.3 Terminals. Battery terminal connections (plus and minus) shall be either:

a. Post or flag type terminals of the same composition with threaded nut, or

b. Receptacle type terminals with threaded insert termination on each of the terminals

The same battery model shall have a unique part number and National Stock Number (NSN) depending on what type of terminal is used. For example, two identical batteries from the same manufacturer but with different terminal connections shall have independent part numbers and NSNs. An epoxy is used as sealing material surrounding the terminals. The epoxy shall guarantee leak-free operation.

2.4 Pressure relief valve. The battery shall have sufficient pressure relief valves to prevent rupturing of the battery when internal pressure exceeds the battery design pressure. The relief valve shall be a one-way valve. Vent release pressure shall be between 1 and 2 pounds per square inch (between 6.89 and 13.8 kilopascals). The seal ring material shall be neoprene rubber. The valve shall close when pressure has regulated to prevent entry of outside air into the cell.

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 6140

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2.5 Off-gassing.

a. The VRLA battery shall be designed to provide up to 99 percent recombination under normal charging (see 6.5.2) conditions. As a result of this design, the VRLA battery is often utilized in occupied areas without concern for the hazardous buildup of hydrogen gas. Critical to minimizing the gas emitted from the VRLA battery is the use of the proper float charging voltage (see 2.9) and limited use of the equalization voltage of 2.4 volts per cell.

b. Off-gassing rates are primarily determined by two factors, charging voltage and battery temperature. As the charging voltage per cell is increased above 2.3 volts per cell, the gassing rate increases dramatically. The gassing rate at 2.3 volts per cell is approximately 0.00000065 cubic feet per hour (ft^3/hour) (0.0185 cubic centimeters per hour [cc/hour]) per Ah per cell. However, this rate could increase by a factor of 20 or more at 2.5 volts per cell. The equalization voltage shall be limited to 2.4 volts per cell and the VRLA battery float voltage should be limited to 77 °F (25 °C).

c. Note that the gassing rates are so small that they are given in cc/hour; for reference, there are 28,317 cc per ft^3.

d. The float-charge gassing rate for one battery shall be less than 0.000035 ft^3/hour (1 cc/hour).

e. The accelerated gassing rate for one battery shall be less than or equal to 0.000070 ft^3/hour (2 cc/hour).

2.6 Capacity (see 6.5.1).

a. At a 20-hour rate, the battery capacity shall operate at least 2.75 amperes direct current (DC) to a cut-off voltage of 10.5 volts DC, providing 50 Ah in the temperature range of -4 to 122 °F (-20 to 50 °C).

b. At a 8-hour rate, the battery capacity shall operate at least 6.52 amperes DC to a cut-off voltage of 10.5 volts DC, providing 53.6 Ah in the temperature range of -4 to 122 °F (-20 to 50 °C).

2.7 Reserve capacity (see 6.5.6). The battery shall deliver 33 amperes for a minimum of 60 minutes at 77 °F (25 °C) with a cut-off voltage of 10.2 volts DC.

2.8 Retention of charge (see 6.5.7). The battery shall retain at least 80 percent of the 20-hour-rate capacity (50 Ah) 28 days after charge termination when discharged at the 20-hour rate.

2.9 Cycling. The battery cycling capability shall be as follows. Only cycles yielding 80 percent or greater capacity are to be counted towards the following:

a. The battery shall be capable of a minimum of 150 cycles of:
   (1) 100 percent discharges with a cut-off voltage of 10.5 volts DC, or
   (2) Discharges to 80 percent capacity at the 20-hour rate

b. The battery shall be capable of a minimum of 350 cycles of:
   (1) Discharges never exceeding a 50 percent capacity with a cut-off voltage of 11.55 volts DC, or
   (2) Discharges to 80 percent capacity at the 20-hour rate

c. The battery shall be capable of a minimum of 800 cycles of:
   (1) Discharges never exceeding a 30 percent capacity with a cut-off voltage of 11.87 volts DC, or
   (2) Discharges to 80 percent capacity at the 20-hour rate

Charging for cyclic/Uninterruptible Power Supply applications shall apply a constant voltage charge between 14.1 and 14.7 volts DC at 68 °F (20 °C), with the initial charging current not to exceed 0.25C (see 6.5.3). Charging for float or service applications only shall apply a constant voltage charge between 13.5 and 13.87 volts DC at 68 °F (20 °C), with the initial charging current not to exceed 0.25C.

2.10 Lifting procedure. Each battery shall be provided with a lifting procedure to facilitate manual or mechanical handling.

2.11 Ripple voltage. The battery shall be able to handle a maximum ripple voltage, as a percentage of the applied battery float voltage, up to 5 percent peak-to-peak (3.5 percent of the root mean square [rms] voltage) during continuous operation without derating the published battery life capacity.

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2.12 Ripple current.

a. The maximum rms ripple current shall not exceed 5 percent of the 8-hour rated Ah capacity.

b. The maximum ripple current is often expressed as 5 amperes per 100 Ah of the 8-hour rated capacity.

2.13 Equalizing charge capability (see 6.5.5). The battery shall be capable of handling an equalizing charge every 3 months or every 20 discharge cycles. The battery shall be capable of handling a maximum current of 0.3C with a charge time between 12 and 24 hours below 77 °F (25 °C) without decreasing rated published life capacity.

2.14 Flame-retardant battery case. All batteries are required to have a flame-retardant case. The battery case construction shall utilize high tensile acrylonitrile butadiene styrene (ABS) plastic and comply with UL94 rating between V-0 and V-2.

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2 Safety. A Safety Data Sheet is required in accordance with 29 CFR 1910.1200 and FED-STD-313.

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Code of Federal Regulations (CFR). CFR parts are available online at www.ecfr.gov.

6.1.2 Federal standards. Federal standards are available online at https://quicksearch.dla.mil.

6.1.3 Underwriters Laboratories, Inc. (UL) standards. UL standards are available online at www.ul.com.

6.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision.

b. Product conformance provisions.

c. Packaging requirements.

d. Quantity required.

6.3 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in FED-STD-376 and all other requirements of this CID are met.

6.4 Potential sources. The products in table I comply with the salient characteristics of this CID. Competition is not limited to the indicated sources.

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TABLE I. Potential sources of compliant batteries.

Manufacturer	Manufacturer's battery code
Power Sonic Corp. 7550 Panasonic Way San Diego, CA 92154	PHR-12200
C&D Technologies, Inc. 1400 Union Meeting Road Blue Bell, PA 19422-0858	UPS12-210MR NSN: 6140-01-529-8706
NPPower International Inc. 1845 S Vineyard Ave Unit #2 Ontario, CA 91761	HR12200W

6.5 Definitions.

6.5.1 Capacity. The total amount of energy a battery can store and produce through electrochemical reactions. Battery capacity is usually measured in Ah but can also be measured in watt-hours or milliampere-hours.

6.5.2 Charging. The process of converting chemical energy into electrical energy in a battery cell or pack.

6.5.3 C-rate. The current that discharges the battery in 1 hour. When C-rate is displayed in the context of charging, it is presented with a number and C, (*x*C). This number is a percentage which is applied to the 1-hour discharge value of the battery. Example: If a battery can discharge 100 amps for 1 hour, and has a charge rating of 0.25C, this indicates the battery is capable of charging at 25 amps continuously depending on state of charge.

6.5.4 Discharge. The process by which a battery loses energy or voltage, releasing stored electrical energy to power a device or system.

6.5.5 Equalization charge. A charging cycle that intentionally overcharges the battery to prevent and partially reverse sulfation (see 6.5.8). This overcharge is minimal.

6.5.6 Reserve capacity. A measurement of how long a fully charged battery can supply a specific amount of power before its voltage drops below a certain level.

6.5.7 Retention of charge. A battery's ability to maintain its capacity over time, as a percentage of its initial capacity.

6.5.8 Sulfation. The process of the lead-acid within a battery reacting with sulfate to crystalize and form lead-sulfate. Lead-sulfate will permanently reduce the capacity of a battery if not addressed. The process of sulfation occurs naturally in VRLA batteries.

6.6 Key words.

Fire Retardant (FR)
Lowest replaceable unit (LRU)
Rechargeable
Secondary battery
UPS battery

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CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army – CR
Navy – SH
Air Force – 99

Review activities:
Army – AT, AV, MI
Navy – AS
Air Force – 20, 71

Preparing activity:
Navy – SH
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